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MCN201 SUSTAINABLE ENGINEERING

MODULE 1

Sustainability: Introduction, concept, evolution of the concept; Social, environmental and economic sustainability concepts; Sustainable development, Nexus between Technology and Sustainable development; Millennium Development Goals (MDGs) and Sustainable Development Goals (SDGs), Clean Development Mechanism (CDM).

1.1 WHAT IS SUSTAINABILITY?

The simple definition that sustainability is something that improves **"the quality of human life while living within the carrying capacity of supporting eco-systems"** In environmental science it can be defined as the quality of not being harmful to the environment or depleting natural resources, and thereby supporting long-term ecological balance.

"The ability to sustain something for an indefinite period of time, without depleting the resources used to sustain it, and such that it does not damage the surroundings (environment) in which it resides."

Examples: Long-lived and healthy wetlands and forests are examples of sustainable biological systems

Specific types of sustainability include,

- sustainable agriculture,
- sustainable architecture
- Ecological economics

1.2 NEED OF SUSTAINABILITY

The need of sustainable environment is related with the need of protection of the environment at large which is a source of everything. Some of the need of sustainable environment is;

- Environmental destruction is a direct or indirect threat to the right to life and livelihood.
- Environment (natural resources) can be a source of conflict that might lead to human rights violations.

1.3 HUMAN, SOCIAL, ECONOMIC AND ENVIRONMENTAL SUSTAINABILITY

1.3.1 Comparison of Human, Social, Economic and Environmental Sustainability: **Human Sustainability**

- Human sustainability means **maintaining human capital**.
 - Human capital is a private good of individuals, rather than between individuals or societies.
 - The health
 - Education
 - Skills
 - Knowledge & leadership
- As human life-span is relatively short and finite (unlike institutions) **human sustainability needs continual maintenance** by investments throughout one's lifetime
- **Promoting maternal health and nutrition, safe birthing and infant and early childhood care fosters the start of human sustainability.** Human sustainability needs 2–3 decades of investment in education and apprenticeship to realize some of the potential that each individual contains. Adult education and skills acquisition, preventive and curative health care may equal or exceed formal education costs

1.3.2 Comparison of Human, Social, Economic and Environmental Sustainability: **Social Sustainability**

- Social sustainability means **maintaining social capital**. Social capital is investments and services that create the basic framework for society. It **lowers the cost of working together** and facilitates cooperation: trust lowers transaction costs. Only systematic community participation and strong civil society, including government can achieve this.
 - Cohesion of community for mutual benefit
 - connectedness between groups of people
 - reciprocity
 - compassion
 - patience
 - forbearance

- Fellowship, love, discipline and ethics.
- Commonly shared rules, laws, and information (libraries, film, and diskettes) promote social sustainability

1.3.3 Comparison of Human, Social, Economic and Environmental Sustainability: **Economic Sustainability**

- **Economic capital** should be maintained. The widely accepted definition of economic sustainability is **maintenance of capital, or keeping capital intact**.
- Economic and manufactured capital is substitutable. There is much overcapitalization of manufactured capital, such as too many fishing boats and sawmills chasing **declining** fish stocks and forests

1.3.4 Comparison of Human, Social, Economic and Environmental Sustainability: **Environmental Sustainability (ES)**

- Although ES is needed by humans and originated because of social concerns, ES itself seeks to improve human welfare by protecting Natural Capital (NC). As contrasted with economic capital, **NC consists of water, land, air, minerals and ecosystem services**; hence much is converted to manufactured or economic capital. **Environment includes the sources of raw materials used for human needs**, and ensuring that sink capacities recycling human wastes are not exceeded, in order to prevent harm to humans
- Humanity must learn to live within the limitations of the biophysical environment. ES means NC must be maintained
- Technology can promote or demote ES. Non-renewable cannot be made sustainable, but quasi-ES can be approached for non-renewable by holding their depletion rates equal to the rate at which renewable substitutes are created. There are no substitutes for most environmental services, and there is much irreversibility if they are damaged

1.4 SUSTAINABLE DEVELOPMENT

“Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs”

The overall goal of sustainable development (SD) is the long-term stability of the economy and environment; this is only achievable through the integration and acknowledgement of

economic, environmental, and social concerns throughout the decision making process. Sustainable development includes the four interconnected domains:

1. Ecology
2. Economics
3. Politics and
4. Culture.

Sustainable development should have the following features:-

1. Satisfying human needs
2. Favouring a good quality of life through decent standards of living
3. Sharing resources between rich and poor
4. Acting with concern for future generations
5. Looking at the 'cradle-to-grave' impact when consuming
6. Minimizing resource use, waste and pollution

1.4.1 MEASURES OF SUSTAINABLE DEVELOPMENT

❖ Technology

- Using appropriate technology is one which is locally adaptable, eco-friendly, cost effective, resource efficient and culturally suitable.
- Nature is often taken as a model, using the natural conditions of that region as its components.
- This concept is known as “design with nature”.

❖ Reduce, Reuse, and Recycle Approach:

- The 3-R approach advocating minimization of resource use, using them again, and recycling the materials.
- It reduces pressure on our resources as well as reduces waste generation and pollution.

❖ Promoting Environmental Education and Awareness:

- From the childhood onwards, we should develop a feeling of belongingness to our planet.
- This is possible only by introducing environment as a subject in education from the primary level itself.

❖ **Resource Utilization as Per Carrying Capacity:**

- Any system can sustain a limited number of organisms on a long-term basis which is known as its carrying capacity.
- If the carrying capacity of a system is crossed environmental degradation starts.

❖ **Improving Quality of Life Including Social, Cultural and Economic Dimensions:**

- Development should not focus just on one-section of already affluent people.
- Rather it should include sharing of benefits between the rich and the poor.
- The tribal, ethnic people and their cultural heritage should also be conserved

1.4.2 NEXUS BETWEEN TECHNOLOGY AND SUSTAINABLE DEVELOPMENT

❖ Technology is the offspring of science. Technological innovation can be seen as a ‘double edged sword’, with respect to sustainable development.

❖ Technology improves quality of life, eliminate diseases and increase life expectancy

❖ On the other hand, technology creates irreparable environmental damage due to resource extraction and pollution of air, water, soil.

❖ As technology advances, the environmental degradation accelerates exponentially.

❖ Also the benefits of technological innovations are mostly enjoyed by the developed countries.

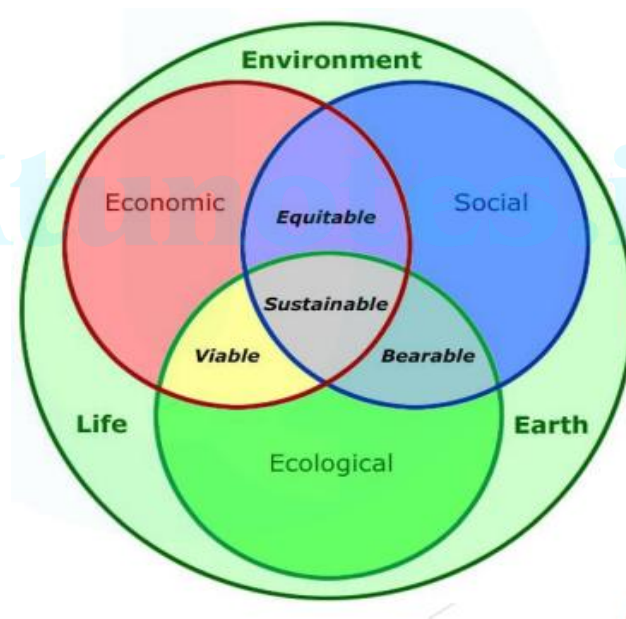
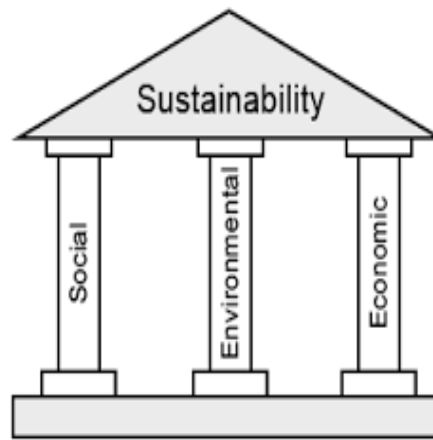
❖ The technology remains as a dream for underdeveloped countries which still face poverty, inadequate sanitation facilities etc.

❖ Hence it is essential to integrate technology, society into sustainability.

❖ Technology can support sustainability by

- conserving natural capital (renewable and non-renewable resources)
- Reducing waste and pollution
- raising efficiency standards
- finding substitutes for toxic/hazardous materials

1.5 PILLARS OF SUSTAINABILITY



1.5.1 ECONOMIC SUSTAINABILITY: means.....

1. Launch program to reduce automobile use
2. Establish a modern bus mass transit scheme
3. Make bus transit fast, cheap and comfortable

4. Place high density living near major arterials
5. Zone for mixed residential/commercial use
6. Expand green zones to safeguard open space
7. Enlarge the amount of per capita green space
8. Enact regulations to protect every urban tree
9. Allow poor to swap their garbage for food
10. Encourage residents to separate their garbage
11. Set up programs to recycle recyclable
12. Enlist the aid of children in recycling efforts
13. Develop a low emissions industrial zone
14. Enact policies to give the poor basic services
15. Give poor free medical and dental care
16. Give poor free child care so they can work

1.5.2 SOCIAL SUSTAINABILITY: means...

1. View natural resources as limited in nature
2. Cultivate the lushness of the settlement area
3. Stress equitable distribution over production
4. Rely on information, not machinery
5. Establish deliberative decision process
6. Value family/community over individuals
7. Work for enjoyment rather than avoiding toil

8. Cherish folk life rather than entertainment
9. Reduce family size and resource use
10. Eliminate divisions of clan, caste, class
11. Practice gender-neutral opportunity policies
12. Strive for universal education of population
13. Address disparities in economic attainment
14. Level the economic playing field for all
15. Make all citizens economic stake-holders
16. Subsidize food, health care, and education
17. Work deliberately to use resources efficiently
18. Address 'wellness needs' of the population
19. Meet 'wellness needs' on an all-for-one basis

1.5.3 ENVIRONMENTAL SUSTAINABILITY

1. Propose a plan to protect natural systems
2. Form team of indigenous resource managers
3. Educate the team in environmental planning
4. Survey the landscape's natural attributes
5. Identify natural opportunities and constraints
6. Identify sensitivities of plants and animals
7. Identify social opportunities/constraints
8. Identify cultural opportunities/constraints

9. Apply eco-principles from other regions
10. Adapt environmental laws from other regions
11. Draft a nature friendly development plan
12. Establish nature reserves and protected areas
13. Establish environmental protection council
14. Provide 'one-stop' development permission
15. Establish community participation committee
16. Hear local citizens affected by development
17. Host democratic fore of citizen participation

1.6 TECHNOLOGY AND SUSTAINABLE DEVELOPMENT

The Goals of sustainability is

- To feed, nurture, house, educate and employ growing human population, *while*
- Conserving earth's basic life support systems and biodiversity *and*
- Reducing hunger and poverty.

Specific challenges in the “**WEHAB**” areas are

- Water, Energy, Health, Agriculture, Biodiversity

1.6.1 NEW TECHNOLOGIES FOR SUSTAINABLE TRANSPORT

The current trends in the transportation system are not meeting the sustainability requirements. It causes large amount of air pollution and thereby causing environmental and health problems. The modern technologies will reduce the carbon emission from transportation sector

Three different technologies to promote sustainable development in the transport sector are :

1. ALTERNATIVE FUEL VEHICLES (AFV)
2. ADVANCED TECHNOLOGY VEHICLES (ATV)
3. INTELLIGENT TRANSPORT SYSTEMS (ITS)

1. ALTERNATIVE FUEL VEHICLES

Alternative fuels are being used today in place of gasoline and diesel fuel made from petroleum e.g. biodiesel, electricity, ethanol, hydrogen, methanol, natural gas, propane This alternative fuels reduces the CO₂ emission.

2. ADVANCED TECHNOLOGY VEHICLES

Electric car, hybrid car, hybrid electric vehicles are some examples of advanced vehicles. They reduce the fuel consumption and cause less pollution to the atmosphere

3. INTELLIGENT TRANSPORT SYSTEMS

More sustainable transport can be achieved through the use of information technology for the Management of transports, so called Intelligent Transport Systems (ITS)

1.6.2 TECHNOLOGY FOR SUSTAINABLE INDUSTRIAL PRODUCTION

Industrial production process starts from raw material selection and ends at consumer goods. Many pollutants are emitted in this process from different areas of production. The modern technologies used in industry are

1. INDUSTRIAL BIOTECHNOLOGY
2. IT FOR RESOURCE MANAGEMENT
3. CLEANER TECHNOLOGY FOR WASTE MANAGEMENT

1. INDUSTRIAL BIO-TECHNOLOGY

It is based on the biological actions. It focuses on the molecular structures and mechanisms, genetic basis and ecology of all living things. This causes reduction in raw material and energy consumption, as well as less pollution and recyclable and biodegradable waste, for the same level of production

2. IT FOR RESOURCE MANAGEMENT

The use of information technology helps in maintaining the production process in most efficient way .IT helps in the LCA analysis of the product which reduces the raw material usage , energy consumption, ect.

3. CLEANER TECHNOLOGY FOR WASTE MANAGEMENT

Examples of new technologies are pre-treatment facilities for combustible waste or organic waste, new large-scale technologies for treating organic waste (e.g. anaerobic digestion), source separation systems and central separation facilities.

Old methods are incineration and land filling

Recycling or reusing is the another method for waste minimization

1.6.3 TECHNOLOGY FOR SUSTAINABLE AGRICULTURE

Agricultural sustainability is necessary for sufficient and safe food to meet the requirements of growing population. Modern technologies are

- 1. Organic farming:** Organic farming can be defined as an approach to agriculture with self-regulating agro-ecosystems, locally or farm-derived renewable resources and the management of ecological and biological processes and interactions. It uses organic fertilizers and organic methods without affecting the eco system.

Quality of Life Concerns	Economic Issue		Social Issue		Environmental Issue	
	Unsustainable	Sustainable	Unsustainable	Sustainable	Unsustainable	Sustainable
Water	High cost of drinking water	Drinking water availability at low cost	Access to drinking water denied to weaker section	Adequate water supply to all sections	High-level of pollution in lakes and rivers	Conservation of existing fresh water bodies
Food	High cost of food and use of fertilizers and pesticides in farming	Good food available at low cost	Access to good food denied to weaker section	Adequate access to good food to all sections	Overuse of fertilizers and pesticides pollute the environment. Deforestation - conversion of forestland to farms	Food is of nutritious quality and related diseases are lowered.
Energy	High cost and intermittent power supply	Electricity available at low cost	Over use of energy by the rich society and inadequate energy distribution	Adequate energy available to all sections	Use of fossil fuels and pollution	Use of renewable resources.(solar, wind, biomass)

1.7 CHALLENGES OF SUSTAINABLE DEVELOPMENT

- First, the **world's population is growing rapidly** and most of this growth will take place in cities. Cities are particularly vulnerable to environmental challenges due to their high population density.
- Second, **increased demand and competition for scarce resources** such as energy, water and food will put upward pressure on resource prices.

- Third, **climate change** and its potential impact

The path of sustainable development can only lead us to the future. Some of the major challenges exist in the modern world that may hinder the effort towards sustainable development are given below:-

1. **Misconception about sustainable development** The popular perception that development and environment protection should not go together, has to be changed.
2. **Population explosion** in developing economies (require more food, energy, goods)
3. **Over exploitation of natural resources** in developed countries (resulting in imbalance between developed & underdeveloped countries)
4. **Poverty** (To achieve sustainable development, eradication of poverty is vital.
5. **Over dependence on fossil fuels** leading to global warming & climate change
6. **Loss in Biodiversity**
7. **Freshwater scarcity**
8. **High cost of appropriate technology** to tap renewable energy sources
9. **Poor solid waste management system** (focusing on mere waste disposal)
10. **Public awareness**
11. **Absence of strict environment laws** and legislations to control pollution
12. Absence of adequate political and industrial will for moving towards a sustainable future
13. **Lack of co-ordination** between the three pillars of sustainable development – Economy, Society, Environment – in finding long term solutions
14. Corruption and misuse of sustainable development assistance funds

1.8 MILLENIUM DEVELOPMENT GOALS

1. To eradicate extreme poverty and hunger
2. To achieve universal primary education
3. To promote gender equality and empower women
4. To reduce child mortality

5. To improve maternal health
6. To combat HIV/AIDS ,malaria and other diseases
7. To ensure environmental sustainability
8. To develop a global partnership for development

1.9 SUSTAINABLE DEVELOPMENT GOALS (SDGs)

The 17 sustainable development goals (SDGs) to transform our world:

- GOAL 1: No Poverty
- GOAL 2: Zero Hunger
- GOAL 3: Good Health and Well-being
- GOAL 4: Quality Education
- GOAL 5: Gender Equality
- GOAL 6: Clean Water and Sanitation
- GOAL 7: Affordable and Clean Energy
- GOAL 8: Decent Work and Economic Growth
- GOAL 9: Industry, Innovation and Infrastructure
- GOAL 10: Reduced Inequality
- GOAL 11: Sustainable Cities and Communities
- GOAL 12: Responsible Consumption and Production
- GOAL 13: Climate Action
- GOAL 14: Life Below Water
- GOAL 15: Life on Land
- GOAL 16: Peace and Justice Strong Institutions
- GOAL 17: Partnerships to achieve the Goal

1.10 CLEAN DEVELOPMENT MECHANISM (CDM)

- The Clean Development Mechanism is regarded as one of the most important internationally implemented market based mechanisms **to reduce carbon emissions**.
- Created under the Kyoto Protocol, the CDM was designed to help developed nations meet **domestic greenhouse gas (GHG) reduction** commitments by investing in low-cost emission reduction projects in developing countries.
- The Clean Development Mechanism (CDM), established under the Kyoto Protocol, is the primary international offset program in existence today.
- It generates offset through investments in GHG reduction, and avoidance projects in developing countries.
- These offset credits, called Certified Emission Reduction credits (CERs), represent a reduction in one metric ton of carbon dioxide (CO₂) emitted to the atmosphere.
- Developed countries can use CERs to more cost effectively achieve their Kyoto Protocol GHG emission reduction targets.
- The stated purpose of the Clean Development Mechanism is to help developing countries achieve sustainable development, and assist industrialized countries in complying with their emission reduction commitments.

1.10.1 PURPOSE OF CLEAN DEVELOPMENT MECHANISM

- Private companies fund projects in developing countries that reduce greenhouse gas emissions.
- They must also meet sustainable development criteria and the “additionality” requirement, which means the emission reductions made, must be “additional” to what would have been possible without CDM funding.
- Upon verification, the CDM awards these projects certified emission reductions (CERs), each equivalent to one ton of carbon dioxide.
- CERs are then sold to developed countries, which use them to meet a part of their reduction commitments under the Kyoto Protocol.
- CERs are also called “offset credits” because they “offset” the developed countries’ emissions with reductions in developing countries.
- CDM allows countries to continue emitting green house gases, so long as they pay for reductions made elsewhere.

- The justification for this is based on the premise that it would be far more expensive to implement emission reduction in industrialized countries than in developing countries.
- It would help developing countries to gain sustainable development benefits from the entry of “clean” and more energy efficient technologies

